Korea Technology and Information Promotion Agency for SMEs

(max 10 pages)

Technology Proposal for 2026 TIPA* International Joint R&D Program

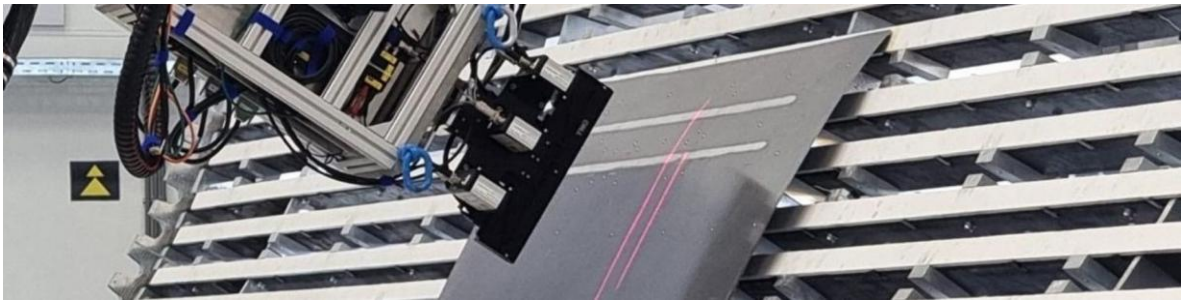
(TIPA: Korea Technology and Information Promotion Agency for SMEs)

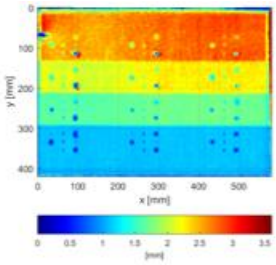
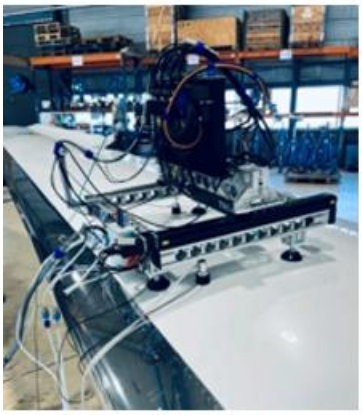
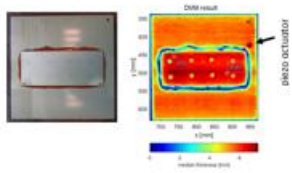
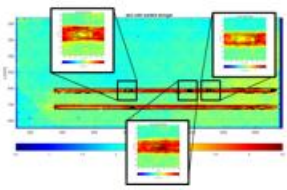
The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

Proposer	Name	Mark Courage	Email	mark.courage@tno.nl
Institute Name	**TNO** Nederlandse Organisatie voor toegepast-natuurwetenschappelijk onderzoek TNO (Netherlands Organisation for applied scientific research TNO)			
Address	Anna van Buerenplein 1, 2595 DA The Hague, The Netherlands			

2. Technology Overview

Title of Proposed Technology	**Advanced Robotics DVM NDT: Direct Velocity Mapping for Advanced Non-Destructive Inspection of Composite and Metal-Metal Connections** 			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display (√), Secondary Battery ( ), Advanced Mobility (√), Next-gen Nuclear Power (√), Advanced Bio ( ), Aerospace/Marine (√), Hydrogen ( ), Cybersecurity ( ), AI (√), Next-gen Communication ( ), Advanced Robotics & Manufacturing (√), Quantum ( )			
Industrial Technology Classification [1]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Mechanical/materials		*Electrical/Electronics*	
	Medium Category	Small Category	Medium Category	Small Category
	Robotics/Automation Machinery	*Assembly/Precision Transfer Technologies, Measurement/Sensor Technologies for Automation*	*Measuring Instruments*	*Measurement Sensors and Components*

Concept of Proposed Technology	TNO's DVM NDT (Direct Velocity Mapping Non-Destructive Testing) is a state-of-the-art inspection technology that utilizes MEMS (Micro-Electro-Mechanical Systems) sensors and advanced acoustic techniques to perform high-resolution, non-destructive testing of composite materials and metal-metal joints. DVM NDT enables automated, inline quality control by mapping the propagation velocity of acoustic waves through materials, allowing for the precise detection of internal defects, delaminations, and bonding quality in both composites and metallic connections—without damaging or disassembling the product. The system's MEMS-based acoustic transducer arrays, combined with AI-driven data analysis and 3D reconstruction, provide unparalleled sensitivity and speed for industrial inspection of composites and metal-metal connections.    
Rationale for Joint R&D Project	The growing use of composites and advanced metal joining in high-tech industries (e.g., aerospace, automotive, electronics) demands reliable, scalable, and automated inspection solutions. Traditional NDT methods are often slow, manual, or lack the required resolution for modern materials. DVM NDT addresses these challenges by offering fast, automated, and highly accurate inspection for both composite and metal-metal connections. Joint R&D is needed to adapt and integrate this technology into Korean SME manufacturing environments, combining TNO's expertise in MEMS, acoustics, and AI with Korean partners' strengths in automation, robotics, and market access.
Key Words	1. Advanced Robotics 2. Quality Control 3. Manufacturing Efficiency 4. Non Destructive Testing 5. Composite quality control 6. Metal-metal bonding quality control

3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	Start of the Project	*TRL 4 (Lab validation of integrated modules)*	End of the Project	*TRL 6 (Pilot demonstration in SME environment: Production & performance evaluation of prototypes at a pilot scale)*
R&D Objectives	**Key R&D objectves:** **1.** Develop and optimize MEMS-based Direct Velocity Mapping modules for automated, inline inspection of composite and metal-metal connections **2.** Integrate advanced acoustic imaging and 3D reconstruction for defect detection and dimensional measurement **3.** Implement AI-driven data analysis for real-time quality assessment and process feedback **4.** Demonstrate the technology in a pilot line at a Korean SME The overarching R&D objective of this joint project is to mature TNO's Direct Velocity Mapping (DVM) NDT technology from a laboratory-validated concept into an operationally deployable inspection solution that can reliably support **composite and metal–metal bond inspection within Korean SME manufacturing environments**. Building on the findings from TNO's, the project aims to create a next-generation DVM platform that offers **higher robustness, enhanced defect detectability, improved ergonomics and increased automation**. Central goals include the optimization of MEMS-based acoustic sensor arrays, refinement of multi-frequency Lamb-wave based imaging, and integration of advanced signal processing and AI-based interpretation tools to achieve robust, operator-independent thickness and defect mapping across a broad range of industrial components. A core objective is to **co-develop detailed functional objectives together with the participating Korean SMEs**, ensuring that the DVM system precisely addresses their priority inspection challenges: high-mix production, curved-surface inspection, inline quality control, complex structural geometries, lightweight mobility requirements, and stringent throughput expectations. Therefore, the R&D objectives will be **progressively refined and detailed** based on the concrete use-cases, part geometries, material stacks, and process constraints provided by the Korean SME partners. To ensure transparency, reproducibility and alignment with industrial system requirements, the complete R&D trajectory will follow a **system-engineering driven approach**, expanding the current prototype into a pre-commercial solution through four structured work-packages: • **WP1 – User Requirements Specification (URS):** translating Korean SME use-cases into measurable DVM performance criteria; • **WP2 – System Requirements Specification (SRS):** defining architecture, component-level specifications, inspection workflows and interfaces; • **WP3 – Prototype Development:** integrating MEMS arrays, actuator modules, advanced imaging, AI analytics, and robotic or guided-mobility options; • **WP4 – Testing & Validation:** conducting laboratory validation and in-situ SME pilots to evaluate accuracy, repeatability, cycle time, robustness, inspection ergonomics, and system throughput. This structured R&D process will allow the consortium to determine the **actual CAPEX/OPEX envelope** and define the business case for industrial adoption, including maintenance demands,			

	operator training, productivity gains, and reduction of scrap and rework. Ultimately, the objective is to deliver a validated, SME-deployable DVM NDT prototype at TRL 6 that demonstrates **clear added value over conventional ultrasonic, eddy-current, or radiographic NDT techniques**, especially in situations requiring fast, automated, operator-independent quality assurance.
R&D Target Specifications	**Key R&D target specifications:** 1. Defect detection accuracy: ≥ 95% (internal and interfacial defects) 2. Inspection cycle time: < 60 seconds per part 3. Adaptability to at least 5 different product types without manual reprogramming 4. Develop prototype of mobile DVM NDT scanner The R&D target specifications aim to translate the project objectives into measurable system-level and component-level technical performance criteria. These targets build upon insights from lab testing, and TNO's latest DVM laboratory performance. They form the initial baseline; however, **final specifications will be further detailed based on the manufacturing use-cases, component geometries, defect types and process flows specified by the Korean SME partners** during the URS phase. **Initial target specifications include:** • **Defect detection accuracy ≥ 95%** for internal wall-thickness deviations, debonds, inclusions, and disbonds within composites and metal–metal joints, including in the presence of curvature or sub-surface structural features. • **Thickness accuracy of ≤ 50 µm**, in line with results demonstrated in the TNO lab and comparable to conventional pulse-echo inspection performance on thin structures. • **Inspection cycle time < 60 seconds** per part (or per scan window), supporting semi-inline and inline quality assurance in SME manufacturing. • **Automated adaptability to ≥ 5 different product types** without reprogramming, reflecting Korean SMEs' high-mix/low-volume production reality. • **Mobile scanning capability**, improving on earlier lab prototype limitations by optimizing weight, ergonomics, stability, and cable-free actuation where feasible. • **AI-enhanced defect classification**, including automated isolation of skin-only thickness even in the presence of ribs/stringers—addressing a key limitation identified in the lab tests on industrial components. • **Robust environmental performance**, specifically improved resilience against temperature shifts, humidity, surface coatings, and vibration—factors shown to influence Lamb-wave behaviour in both laboratory and aviation-MRO environments. • **Seamless integration with robotic or guided manual handling**, based on learnings from the cobot-based lab experiments. These targets will be evaluated and refined throughout the R&D trajectory via the WP1–WP4 system-engineering approach. In WP4, Korean SME pilot implementations will assess performance under real process conditions, enabling reliable calculation of **CAPEX/OPEX, ROI, expected cost savings, and production throughput improvements**. The final specification set will support a validated business case and provide the foundation for potential commercialization and broader Asian market entry.

R&D Plan	High level preliminary planning to be aligned and adjusted with the use cases: 1. **Month 0-6:** Define User Requirement Specification and System requirement specification based on SME use-cases. 2. **Month 6-18:** develop and extend existing lab prototype with System architecture, AI model development, imaging/NDT module integration, robotic handling integration incl. preliminary test runs 3. **Month 18-28:** Pilot demonstration, performance validation, 4. **Month > 28**: technology & commercialization roadmap development and industrial implementation at participating SME's

4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	1. Provide acoustics DVM NDT expertise 2. Lead development of MEMS/acoustic DVM NDT modules; 3. Develop prototype; 4. AI algorithms, and imaging integration; 5. System validation; 6. Knowledge transfer .
Expected R&R of Korean SME Partner	1. Provide manufacturing use cases; 2. Provide materials for measuring on current lab DVM NDT MVP; 3. Adapt and integrate DVM NDT into production lines; 4. Support pilot implementation and market introduction.

5. Expected Outcomes/Impact

Expected Outcomes and Impact	**Technological:** • First-of-its-kind, fully automated, MEMS-enabled Direct Velocity Mapping NDT platform for high-mix, high-tech manufacturing • Enhanced defect detection and process control for composites and metal-metal connections **Commercial:** • Accelerated adoption of smart quality control in Korean SMEs • Reduced scrap, rework, and warranty costs; improved product reliability **Economic**: • Higher productivity, lower inspection costs, improved competitiveness for Korean and European SMEs

6. References

(1) *TNO DVM NDT: https://www.tno.nl/en/digital/smart-manufacturing/manufacturing-throughput-quality/non-destructive-inspection/*

(2) *Scientific literature on MEMS-based acoustics, Direct Velocity Mapping, AI-driven NDT, and automated quality control:*

- ***Core Paper (2023, ASME J NDE):*** *Technical foundation for DVM NDT, MEMS array design, Lamb wave analysis, and defect mapping.*
- ***Conference Work (ECNDT 2023):*** *Expanded application to real-world composite structures with performance metrics and pilot implementations.*
- ***Industry Engagement (2025 Hive):*** *Demonstrated readiness and fit for aviation inspection, paving the way for industrial field trials.*

(3) *Patents:*

- ***PLT2020075 Non-contact inspection technique using MEMs-microphones***
- ***PLT2023031 Non-contact inspection technique for anisotropic composites***
- ***PLT2023046 Patent application 3D nesting***
- ***I-Depot visual robot feedback for tracking and picking***
- ***PLT2017057 User assistance in assembly – operator work station***